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Proceedings of the Combustion Institute 33 (2011) 259–265
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

Dissociation of C_3H_3I and rates for C_3H_3 combination at high temperatures

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Available online 6 August 2010

Abstract

The decomposition of propargyl iodide, $C_3H_3I \rightarrow C_3H_3 + I$, and the following combination of propargyl radical, $C_3H_3 + C_3H_3 \rightarrow C_6H_6$, have been observed in the incident shock in mixtures of 1%, 2%, and 5% C_3H_3I in Kr, over 800–2000 K, for nominal post-shock pressures of 20 and 130 Torr. The iodide dissociates readily, the endothermic reaction producing an easily resolved positive density gradient whose kinetic modeling allows extraction of precise rates showing strong falloff. A simple Gorin-model RRKM treatment of the measurements results in a $k_\infty = 3.35 \times 10^{18}\ T^{-0.9}\exp^{(-22192/T)}\ s^{-1}$. The initial positive gradient is immediately followed by a strong negative, showing exothermic reaction from propargyl combination. Modeling of this gradient with an assumed theoretical product distribution, provided by J.A. Miller, results in propargyl combination rates in excellent agreement with current theory and having a strong negative temperature dependence, the rate dropping near a factor of forty over 800–2000 K. Possible errors associated with the need to assume a product distribution are discussed and it is suggested that this should not result in rate uncertainties for the C_3H_3 combination reaction greater than ±50%.
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Keywords: Propargyl; Soot; Shock tube; Propargyl iodide

1. Introduction

There is now little doubt that the propargyl combination, reaction (1), is a major source of aromatics, PAH and soot in aliphatic flames and pyrolytic systems.

$$C_3H_3 + C_3H_3 \rightarrow \text{isomers of } C_6H_6 \quad (1)$$

It is of particular consequence there because of the resonance-stabilized radical's high stability and low reactivity toward oxygen, allowing its buildup to large concentrations. This reaction has now been extensively analyzed both through experiment [1–11] and theory [12–19], ever since its proposal by Wu and Kern [20] to explain their observation of benzene in shock tube experiments on allene decomposition. It has now been shown to be an exceptionally complex process generating multiple stable products through a variety of paths [17].

Most of the experimental studies of reaction (1) were confined to relatively low temperatures, below 700 K, using photolytic generation of the C_3H_3 radicals, although there are now three shock tube investigations using thermal dissociation of C_3H_3I for radical formation, reaction (2), [2,8,10] that reach temperatures above 1000 K.

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doi:10.1016/j.proci.2010.05.030

$$C_3H_3I \rightarrow C_3H_3 + I \quad (2)$$

Low-T photolytic studies include those of Alkemade and Homann [1], Morter et al. [4], Fahr and Nayak [9], Atkinson and Hudgens [3], DeSain and Taatjes [5], Giri et al. [6] and Shafir et al. [7], who extended the photolytic investigations to 1000 K. Product information is available in Refs. [1,2,7,9] and especially [10]. The low-temperature results are numerous and the present paper is primarily concerned with higher temperatures, so for further details of this work the reader is referred to the complete listing given in Table 1 of Refs. [8,10].

Extensive theoretical work on C_3H_3 combination has been performed over several years by J.A. Miller and co-workers, most recently employing high-level electronic structure determination of energies and structures of both products and transition states [17,19]. These results were then used together with master-equation calculations to provide rate constants including falloff effects for all important channels. The product distribution predicted in this theoretical work was found to be close to the available experimental data.

Here, we are mainly concerned with the higher temperatures previously investigated by Scherer et al. [2], Fernandes et al. [8], and Tang et al. [10] in shock tubes and Rasmussen et al. [11] who modeled C_3H_3 from a well-characterized, low pressure ethylene flame. Rasmussen et al. obtained a rate constant for (1) of 1.3×10^{12} cm^3 mol^{-1} s^{-1} with an uncertainty of a factor of 5.

The first shock tube study is a rather extensive investigation of the dissociation of dilute mixtures of 3-iodopropyne in reflected shocks that covered 1100–2100 K, 1.5–2.2 bar, for initial fractions of 0.6–210 ppm in Ar. They used ARAS detection of H-atoms for the higher temperature, low concentration experiments to set rates for propargyl dissociation (1500–2000 K), and by applying higher concentrations at lower temperatures (1000–1250 K), were able to observe non-resonant propargyl absorption, and follow its loss through reaction (1). They also carried out a residual gas analysis and found several C_6H_6 species, although they were unable to positively identify all of them. However, they did observe a shift from aliphatics to benzene as the temperature increased. Rate constants for (1) were found to lie in the range 4.5×10^{12} – 9×10^{12} cm^3 mol^{-1} s^{-1}. An important additional conclusion from the ARAS results was that reaction (3):

$$C_3H_3 + C_3H_3 \rightarrow C_6H_6 + H \quad (3)$$

contributes less than 5% of (1).

The incident shock experiments of Fernandes et al. [8], also observed the second-order decay of propargyl radicals, but now used absorption at 332.5 nm, in mixtures of 750 and 800 ppm C_3H_3I in Ar, for 995–1440 K and 0.6–1.0 bar. Their analysis produced a rate expression: $k_1 = 5.8 \times 10^{11}\ \exp(3534\ \mathrm{K}/T)$ cm^3 mol^{-1} s^{-1}, in essential agreement with the findings of Scherer et al., but slightly higher than the recommendation of Shafir et al. for 1000 K. Finally Tang et al. [10], employed single-pulse shock tube product analysis at high pressures to determine the product distribution over a range of temperatures. They identified as major products 1,5-hexadiyne (15HD), 1,2-hexadiene-5-yne (12HD5Y), 3,4-dimethylenecyclobutene (34DMB), 2-ethynyl-1, 3-butadiene (2E13BD), fulvene and benzene. Minor products were *cis*- and *trans*-1,3-hexadiene-5-yne (13HD5Y) and 1,2,4,5-hexatetraene (1245HT).

Table 1
Reaction mechanism for the decomposition of C_3H_3I and the association of C_3H_3 at 130 Torr. Unit: cm^3, mol, s and kcal.

Reaction	log A	n	E	$\Delta H_{r,298K}$	Refs.
(1a) $C_3H_3 + C_3H_3 = 15HD$	66.875	−16.59	23.177	−68.68	[a]
(1b) $C_3H_3 + C_3H_3 = 1245HT$	144.58	−38.81	75.705	−73.65	[a]
(1c) $C_3H_3 + C_3H_3 = 34DMCB$	9.199	−0.19	−14.150	−87.91	[a]
(1d) $C_3H_3 + C_3H_3 =$ fulvene	67.981	−16.37	31.197	−116.86	[a]
(1e) $C_3H_3 + C_3H_3 = 12HD5Y$	37.094	−7.95	6.110	−68.54	[a]
(1f) $C_3H_3 + C_3H_3 + M = 2E13BD$	68.073	−16.30	30.044	−85.84	[a]
(1g) $C_3H_3 + C_3H_3 + M =$ benzene	68.186	−16.27	35.149	−148.47	[a]
(1h) $C_3H_3 + C_3H_3 + M = 13HD5Y$	63.968	−15.17	29.375	−86.36	[a]
(1i) $C_3H_3 + C_3H_3 = C_6H_5 + H$	51.913	−11.15	32.602	−33.71	[a]
(2) $C_3H_3I + M = C_3H_3 + I + M$	66.600	−14.53	59.317	44.50	This work
(3) Benzene $= C_6H_5 + H$	108.130	−25.81	181.750	−114.76	[17]
(4) $H + H + M = H_2 + M$	18.000	−1.00	0.000	−104.20	[33]
(5) $C_6H_5 + C_6H_5 = C_{12}H_{10}$	22.508	−2.81	4.789	−117.6	[32]
(6) $C_6H_5 + C_6H_5 = C_6H_6 + C_6H_4$	−2.668	4.57	−5.743	−30.9	[32]
(7) $H + C_3H_3I = HI + C_3H_3$	13.700	0.00	2.000	−25.76	Estimated
(8) p-$C_3H_4 + M = H + C_3H_3 + M$	17.000	0.00	70.000	92.21	[20]
(9) a-$C_3H_4 + M = H + C_3H_3 + M$	17.000	0.00	70.000	90.78	[20]

[a] J.A. Miller, private communication.

A difficulty shared by the first two shock tube studies that offer a rate of (1) is the need to assign an absorption coefficient for the C_3H_3 from the measured absorption at $t = 0$. Here one must assume the iodide decomposition is immediate so the initial concentration of propargyl is that of the loaded C_3H_3I. Both instrumental resolution and possible slow iodide decomposition could affect this assignment, and the iodide rate has not yet been fully established. However, Fernandes et al. [8], do argue rather convincingly that this problem should impact their rate determinations by no more than 30%. Nonetheless, it would seem that an independent measure of the rate of (1), one that also provides accurate rates for the propargyl iodide decomposition, and one that does not require the extraction of an absorption coefficient, may be of value. Here we apply the laser schlieren (LS) technique, which responds to the net exothermicity of (1), and whose resolution also allows extension to even higher temperatures. Furthermore, the LS technique also provides an accurate determination of the rate of the iodide dissociation from the density gradient at $t = 0$.

2. Experimental section

C_3H_3I pyrolysis and C_3H_3 combination was examined by shock-heating mixtures of 1.00% and 2.00% C_3H_3I in krypton (Spectra Gases, 99.997%), observing the resulting density gradients with the laser schlieren (LS) technique. C_3H_3I was prepared from C_3H_3Br and NaI by a Finklestein reaction [10]. All mixtures were prepared manometrically by vaporizing the liquid reagent, collecting the vapor in a 50 L glass bulb, and then adding the appropriate amount of rare gas diluent. Refractivities were 6.367 for Kr [21] and 27.102 for C_3H_3I. The later value was obtained by additivity using appropriate values from Refs. [21,22]. The mixture refractivity was assumed constant throughout an experiment. This assumption is more than sufficiently accurate with mixtures this dilute in Kr even for such a complex pyrolysis as in the current work. Using data from the compilation of Tanzawa and Gardiner [21], and other methods of estimation [22], we suggest a maximum (late) variation of refractivity of less than 2% in the 5% C_3H_3I experiments, a figure of little consequence in any part of the present analysis.

All axial density gradient ($d\rho/dx$) measurements were made behind the incident shock wave using the familiar laser schlieren technique. Details of the LS technique as used herein and other aspects of the experiments and routine calculations are as described at length elsewhere [23–26]. As in previous work the shock waves were initiated by spontaneous rupture of Mylar diaphragms by helium driver gas. Thermodynamic functions in shock calculations and gradient simulations were taken either from the publication of Burcat and Ruscic [27] or were calculated from the results of *ab initio* calculations. The thermodynamic properties and their sources are summarized in the Supplementary information, Table S1.

3. Results

Illustrative examples from the set of 27 experiments examined are shown in Figs. 1 and 2. Figure 1 exemplifies the LS raw data and Fig. 2 displays selected examples of such data presented as semi-log profiles of derived density gradient.

In all cases the overall result is a strong initial positive signal in the raw data, followed by a sharp, accelerating drop usually leading to fairly strong negative values. Here these features are indicative of initial endothermic reaction, from dissociation of the propargyl iodide (2), followed by a late dominance of exothermic reaction arising from the formation of C_6H_6 species by C_3H_3 combination, reaction (1).

The LS gradient is proportional to the net rate of endothermic reaction, so to extract rates for (1) from the later falling or negative gradients requires an independent estimate of the reaction's overall net exothermicity. This requires thermodynamic properties for all product C_6H_6 as well as the distribution of these at each temperature encountered during the process. Because there is a plethora of C_6H_6 species formed, and no adequate experimental data on their distribution is available for present low pressure, high temperature conditions, we have used the results of the latest calculations from the Miller group [17,19, J.A. Miller, private communication] to provide this distribution.

The most accurate and convenient way to introduce the product distribution is through a simulation with the theoretical rates expressed in modified Arrhenius form, selected for pressures near those of the 20 and 130 Torr experiments. The chosen rate expressions are presented in Table 1 reactions (1a–1i). Several aspects of this choice of mechanism are considered further later on.

These rate expressions were then used together with an initial rate for the iodide decomposition in a trial simulation. To produce a fit, only the rates of C_3H_3I dissociation (2) and the total rate of C_3H_3 combination (1) were adjusted. That is, all the tabulated C_3H_3 combination rates were scaled equally, thus preserving the theoretical product distribution. The result of such simulations and adjustments is depicted with the example semi-log gradient data in Fig. 2. Also included in Fig. 2 are results of simulations that illustrate the sensitivity of the results to variation in k_1 and k_2. The rates thus derived for both the iodide

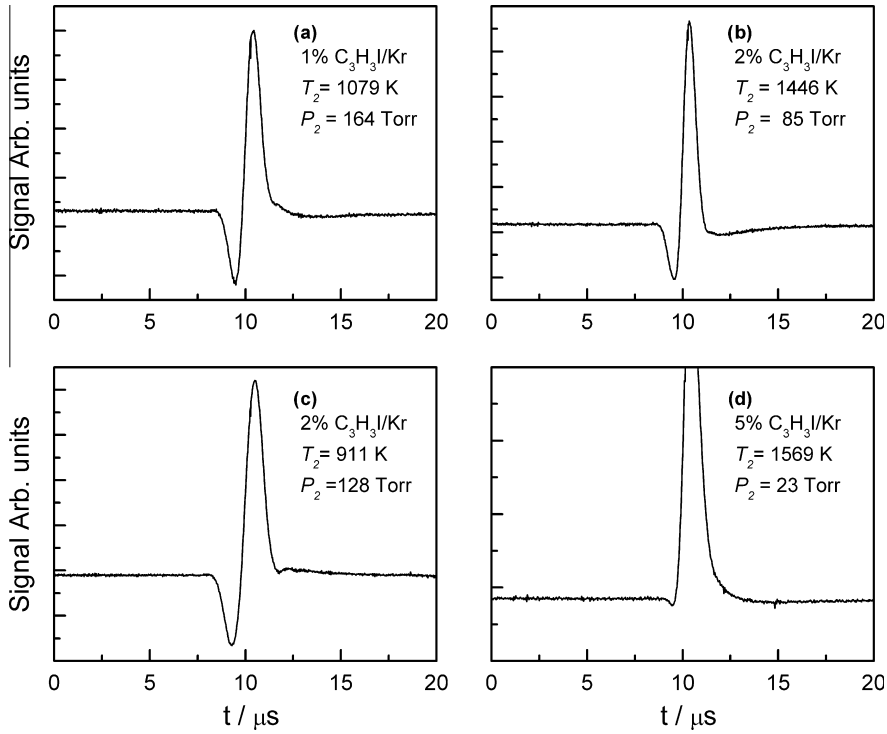

Fig. 1. Example raw (voltage in real time) LS signals. The large spike and preceding valley are due to passage of the shock front through the laser beam and the signal due to reaction follows the large spike. The very small initial dip at the start of the 'chemical' signal, often apparent in LS profiles, and here seen in the bottom left-hand example, is a residual consequence of diffraction and is always ignored.

dissociation and the combination reaction are graphically presented in Fig. 3 for the iodide dissociation and for the combination reaction (1) in Fig. 4 where these are presented together with all the literature proposals. The data along with experimental parameters are tabulated in Table S2, available in the Supplemental information. Figure 3 also contains the result of a simple Gorin-model RRKM [28] fit to the rates derived for (2), showing the strong falloff of the observed rates.

Molecular parameters for the Gorin-model were taken from Burcat and Ruscic [27]. Here $E_0 = 43.3$ kcal/mol from $\Delta H_{r,0K}$ and $\langle \Delta E \rangle_{down} = 500\ cm^{-1}$, similar to that found for other iodide dissociations [29–31], and a hindrance parameter, $\eta = (1 - 0.04 \times T^{0.2})$.

Note that the procedure for the rate of (1) is in no way incestuous. The result is in no way influenced by the magnitude of the rates in Table 1, but only by the distribution they provide, which sets the net exothermicity of reaction (1) to which the LS experiment is sensitive, see Ref. [24] for details. Although during a typical LS experiment the distribution of products does change with time during the negative gradients because the gas is heating itself slightly, this effect is small, and the use of a single product distribution during each experiment would actually be very accurate.

The Miller et al. results [17,19, J.A. Miller, private communication] also include mutual isomerizations of the various product species, but these processes have been ignored herein. The present LS experiments are of very short duration, and the reaction does not proceed very far during the observation period, so that products are in small quantity. Including isomerizations among these alters the gradient by less than 5%, well below the uncertainty in measurement. By removing these steps, all the remaining reactions are consistent in reactants and rate law, so they can then be scaled together. The phenyl radicals generated by (1i), Table 1, are terminated by recombination forming biphenyl ($\Delta H_{r,298} = -117.6$ kcal/mol), disproportionation to benzene + *o*-benzyne ($\Delta H_{r,298} = -30.9$ kcal/mol) [32]. There are three reverse reactions, the decompositions of the linear isomers, that do slightly affect the simulations, but their contribution is very small to start with and becomes smaller as the reaction proceeds, and this issue is again of no real consequence. At the higher temperatures encountered herein, there is the possibility of measurable C_3H_3 dissociation.

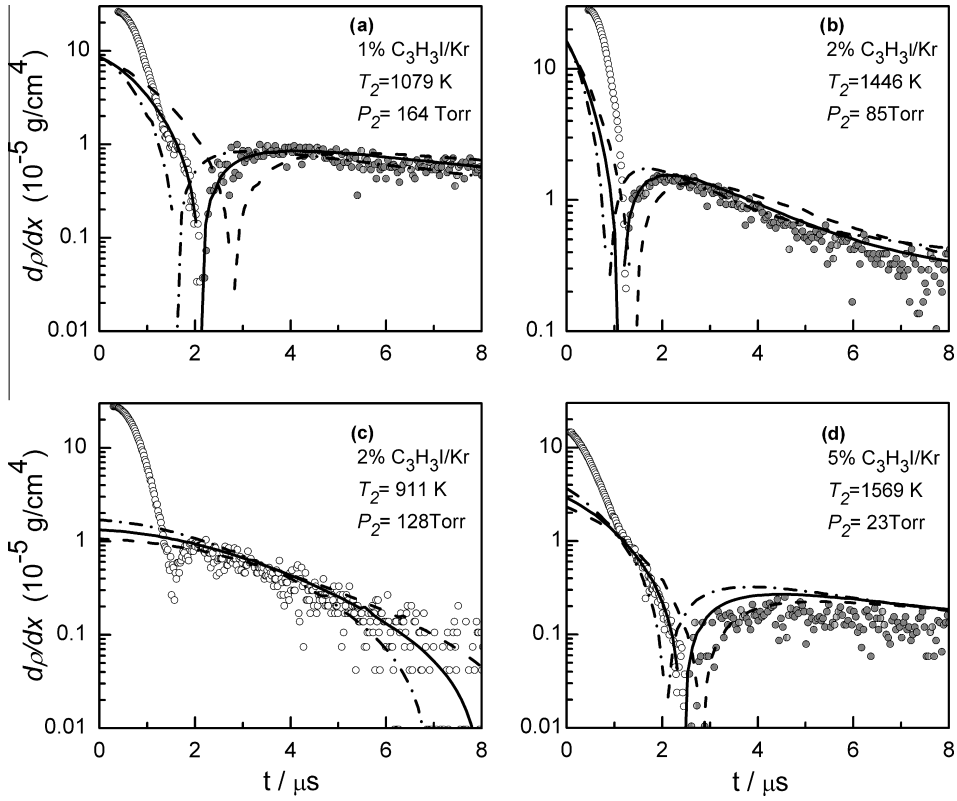

Fig. 2. Semi-log plots of density-gradients derived from laser-schlieren experiments taken from the LS profiles shown in Fig. 1. Points represent absolute values of experimental gradients. Open symbols are positive values while closed symbols represent negative values. The solid black lines are the results of simulations using the model in Table 1. In figures (a) and (b) the dash line and dash-dot line show the effect of altering k_1 by factors of 2 and 0.5, respectively. In figures (c) and (d) the dash line and dash-dot line show the effect of altering k_2 by 0.7 and 1.3, respectively.

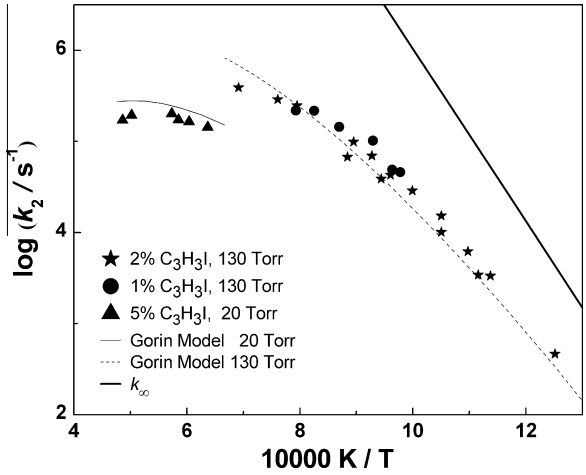

Fig. 3. First order decomposition rates for C_3H_3I. These rates are here compared to a simple Gorin-model RRKM simulation [28] whose molecular parameters are taken from the Burcat database [27]. $E_0 = 43.3$ kcal/mol, $\langle \Delta E \rangle_{down} = 500\ cm^{-1}$, $\eta = 1 - (0.04 \times T^{0.2})$.

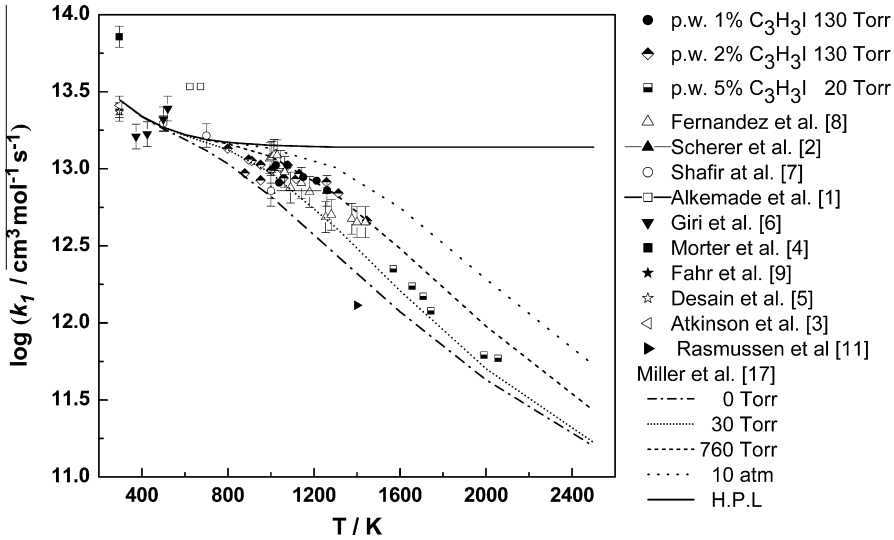

Fig. 4. Derived rate constants (cm^3/mol sec) for reaction (1), the association of C_3H_3, with a comparison to previous experimental work and the theoretical results from Ref. [17].

However, even for the highest temperature accessed herein, 2056 K in a 5% 20 Torr run, this reaction contributes no more than 3% to the gradient at all times.

The results exhibited in Fig. 4 are actually quite close to the theory of Miller and co-workers [19], and accounting for the higher pressures used by Fernandes et al. [8], 600 and 1000 mbar, in excellent agreement with their measurements. They also closely confirm their respective predicted and observed negative temperature dependence. As with the results of Fernandes et al. [8], the present values are high with respect to the photolytic experiments of Shafir et al. [7], around 1000 K, and that of Rasmussen et al. at 1400 K [11] but the discrepancy is now smaller.

In itself the LS experiment is not subject to significant systematic errors [23–26], and as usual, errors introduced in these measurements are best taken as reflected in the scatter, usually below ±20%. This estimate applies to the iodide dissociation rates, but the rates presented here for reaction (1) are also sensitive to the assumed product distribution through the average heat of exothermic reaction in (1), and this is taken from a set of theoretical calculations only partially confirmed by experiment. In addition, there are questions about the accuracy of the heats of formation of the various product C_6H_6 as well as that of the C_3H_3 reactant. However, the situation is not as fraught as it might seem; the range of ΔH^o among the listed reactions in Table 1 contributing significantly to the gradient (d, f, g, and h), is less than a factor 2, and it would take a truly egregious error in the theoretical product distribution to produce anything like such a large error in rate. Thus we propose the rates presented here for (1) should lie well within ±50%.

4. Conclusions

Use of the laser-schlieren technique has permitted the extraction of precise rate data on the overall rate of the exothermic combination of propargyl radicals.

$$C_3H_3 + C_3H_3 \rightarrow \text{isomers of } C_6H_6 \quad (1)$$

over lower pressures and higher temperatures, 800–2000 K, than previous studies. The radicals are formed from rapid C_3H_3I pyrolysis.

$$C_3H_3I \rightarrow C_3H_3 + I \quad (2)$$

in incident shock waves. The density-gradient signals exhibit a large initial positive magnitude, followed by a clear negative signal. The positive gradient results from the C_3H_3I dissociation and the later negative values reflect the progress of reaction (1).

Submicrosecond resolution of the early positive gradient provides accurate rates for reaction (2) over the full temperature range. Reaction (2) shows strong falloff here fit with a simple Gorin-model RRKM calculation. A high-pressure limit rate constant $k_{2,\infty} = 3.35 \times 10^{18}\, T^{-0.9} \exp^{(-22192/T)}\, s^{-1}$ was obtained by model extrapolation. The negative gradients are well described by reaction (1) with a model including thermodynamic properties from calculation and the literature for all the product C_6H_6 species, and a product distribution obtained

from recent high-level theory, when these are combined with an overall rate adjusted to fit. Only the rates of (1) and (2) were adjusted to produce this accurate fit.

The resulting rate constants for (1) are found to be in excellent agreement with theory and with the recent shock-tube experiments of Fernandes et al. [8] when a slight falloff is recognized. The experiments also confirm the strong negative temperature dependence of the high temperature rates.

Acknowledgements

The effort at UIC was supported by the US Department of Energy, Division of Chemical Sciences under Grant No. DE-FE-85ER13384. The work at Argonne was also supported by the agency, but now under Grant No. DE-AC02-06CH11357. We would like to thank J.A. Miller of Sandia National Laboratory for providing the theoretical product distributions and rate coefficient expressions.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.030.

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